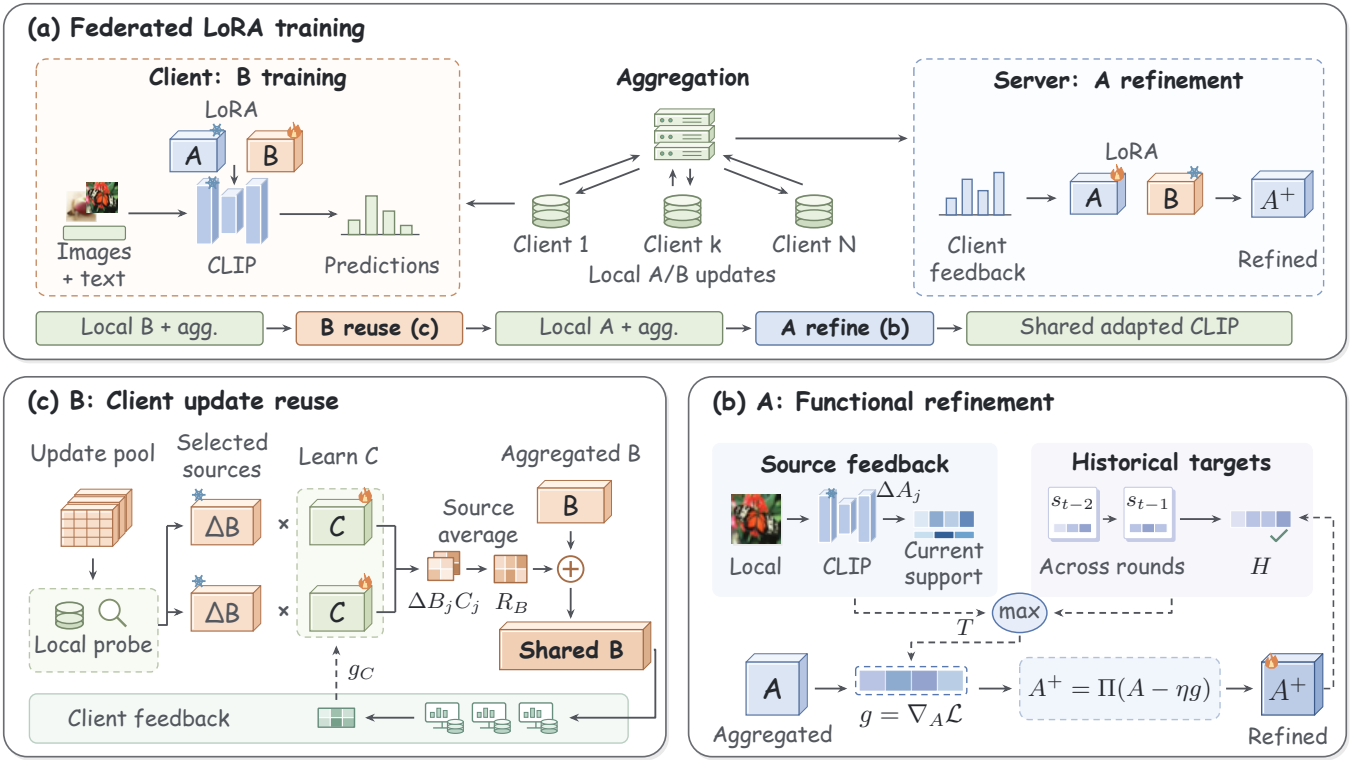

(a) Federated LoRA training
Client: B training
LoRA
A
B
Images + text
CLIP
Predictions
Aggregation
Client 1
Client k
Client N
Local A/B updates
Server: A refinement
LoRA
A
B
A^+
Client feedback
Refined
Local B + agg.
B reuse (c)
Local A + agg.
A refine (b)
Shared adapted CLIP
(c) B: Client update reuse
Update pool
Selected sources
Learn C
Aggregated B
ΔB
ΔB
C
C
Source average
B
$\Delta B_j C_j$
R_B
Local probe
Shared B
g_C
Client feedback
(b) A: Functional refinement
Source feedback
Local
CLIP
ΔA_j
Current support
Historical targets
s_{t-2}
s_{t-1}
Across rounds
H
T
max
A
Aggregated
$g = \nabla_A \mathcal{L}$
$A^+ = \Pi(A - \eta g)$
A^+
Refined
Model flow
Feedback
Frozen
Trainable